

Towards a Unified Modality-Agnostic Multimodal Framework for Cognitive Workload Assessment

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Abstract—Cognitive workload reflects the mental effort required during task performance and is central to the design of adaptive human-machine systems. The use of biosignals to measure cognitive workload has been extensively researched and documented; however, studies examining the effects of combining heterogeneous biosignal modalities for this purpose remain limited. To provide insight into this area, we developed a unified, modality-agnostic, hierarchical Transformer-based architecture to process heterogeneous biosignal modalities within a single model. We use this framework in a pilot study evaluating all 31 possible combinations of five modalities: Electrocardiogram (ECG), Electrodermal Activity (EDA), Respiration (RESP), Peripheral Oxygen Saturation (SpO₂), and Electroencephalogram (EEG), under leave one subject out validation across three cognitively distinct tasks: abstract reasoning (IQ), arithmetic problem solving (MATH), and a game task (GAME). In this pilot setting, the results suggest that: (i) EEG is the strongest single modality, ranking highest in IQ, GAME, and the pooled ALL setting, where samples from all three tasks are combined; (ii) adding more modalities does not consistently improve performance; (iii) the full five modality combination achieves the highest *Average* score of 73.02% on IQ and 68.08% when the *Average* scores are averaged over the four evaluation settings: IQ, MATH, GAME, and ALL; and (iv) the proposed method reduces model size by approximately 50% compared with late fusion alternatives while maintaining lower inference time.

Index Terms—cognitive fatigue, mental workload, physiological signals, brain activity, multimodal fusion, transformer

I. INTRODUCTION

Cognitive Workload is defined as the mental effort required for working memory to complete tasks. In many high-risk fields, including aviation, automotive, robotics, and automation, estimating cognitive workload in real time has significant implications for both safety and performance. The importance of detecting when an individual may require assistance due to increased cognitive workload extends beyond professional set-

tings to health care and education. For example, early detection of increased levels of cognitive workload may provide timely support [1] [2]. Workload assessment methods are broadly classified as subjective or objective [3]. Subjective methods, such as the NASA Task Load Index (NASA-TLX), are well validated but can only be administered after task completion, thereby ruling out real-time monitoring [4]. Objective methods exploit the coupling between mental effort and autonomic and cortical nervous system activity, enabling continuous estimation from physiological sensors [5]. There are several common physiological sensor modalities used for cognitive workload estimation, including EEG for cortical dynamics, ECG for cardiac activity, EDA and respiratory signals for autonomic responses, and blood oxygen saturation for peripheral oxygenation. Each carries distinct and partially complementary information, and no single signal fully characterizes the multi-dimensional nature of cognitive demand [6], [7].

Despite the complementary nature of physiological signals, much of the cognitive-workload literature focuses either on single-modality models or on a limited number of predefined multimodal combinations, rather than systematically comparing the full space of modality subsets [5], [8]–[10]. The lack of systematic evaluation of all modality subsets makes it difficult to provide valid guidance on modality choice when sensor resources are limited, given task variability across application domains [5], [10], [11]. In addition, most multimodal architectures employ modality-specific feature extraction techniques and/or explicit fusion methods, such as intra-modality encoders, dual-branch networks, or decision-level fusion modules, which generally require the sensor set to be defined a priori. Therefore, they are usually designed to process and be optimized for specific input data [12]–[14]. This pilot study addresses these gaps by using a shared hierarchical transformer that processes

any combination of five modalities without modality-specific components, and systematically evaluates all possible modality combinations using a leave-one-subject-out protocol across three cognitively distinct tasks.

II. RELATED WORK

The EEG modality has been studied extensively for cognitive workload estimation, with CNN and LSTM architectures among the most common for extracting spatial and temporal representations from raw or preprocessed signals [15], [16]. However, more complex network designs have shown improved performance [17], and systematic reviews suggest that, in general, deep learning approaches outperform feature engineering pipelines [18]. Transformer-based EEG and brain-activity modeling has also been explored in recent work on affective computing [19]–[21]. Furthermore, Transformer and Conformer architectures have also been applied to EEG-based workload estimation, achieving competitive results at low computational cost [22], [23]. Physiological signals (ECG, EDA, RESP, and SpO₂) provide complementary information that reflects the autonomic nervous system’s responses to cognitive demand. Heart rate variability and electrodermal activity have been shown to discriminate between workload levels during real-world tasks [7], and multimodal wearable datasets have confirmed that ECG and EDA are reliable indicators of stress and cognitive load [3], [24]. The relative contribution of individual peripheral modalities across different cognitive paradigms, however, remains inadequately evaluated. Recent studies have also examined EDA, respiration, ECG, and general biosignal embedding models for physiological-state assessment [25]–[28].

Multimodal physiological fusion has been most thoroughly studied in EEG–fNIRS configurations, where combining cortical electrophysiological activity and hemodynamic responses often improves performance over either modality independently [9], [29]–[31]. Attention-based fusion, mutual information feature selection, and joint feature learning have also been used to improve cross-modal alignment and cognitive state decoding [14], [32]. Related work has further examined heterogeneous physiological fusion, foundation-model designs, compact biosignal representations, facial spatiotemporal representation learning, and co-speech gesture prediction [28], [33]–[37]. In most cases, these methods use modality-specific branches and evaluate a single fixed combination, which limits their applicability when sensor availability varies.

III. METHODOLOGY

A. Data Collection

Eleven participants (6 male and 5 female; ages from 20 to 40 years; mean age 25 ± 5.5) participated in the study. None reported neurological disorders or use of substances affecting the nervous system on the day of the experiment. All participants provided written informed consent, and the study protocol was approved by the Institutional Review Board. EEG was recorded with an *EMOTIV EPOC* wireless headset (14 channels: AF3, F7, F3, FC5, T7, P7, O1, O2, P8, T8, FC6, F4,

F8, AF4) at 128 Hz and preprocessed in EEGLAB, where ICA was used to identify and remove components dominated by ocular or muscular artifacts. Peripheral physiological signals (ECG, EDA, RESP, and SpO₂) were simultaneously acquired with *PLUX Biosignals* sensors at 100 Hz, as illustrated in Fig. 1(a). Participants performed 3 different cognitive tasks. Each of these tasks was presented in 2 difficulty levels. They included a mathematical problem-solving test (MATH), Raven’s Progressive Matrices (for abstract reasoning) (IQ), and an open-source video game (GAME) where participants controlled a ball that bounced around a maze. Task order followed a randomized orthogonal design; each 2-minute activity was followed by a NASA-TLX subjective workload assessment, as depicted in Fig. 1(b), and a 1-minute rest period. Binary class labels correspond to the predefined difficulty level of each task (easy/hard); NASA-TLX scores were collected as a subjective workload verification and are not used as classification labels. The dataset is balanced across both workload classes and all modalities, with each participant providing 12 samples per difficulty level across three tasks, yielding 792 samples per modality and 3,960 samples in total.

B. Multimodal Fusion and Tokenization

Five physiological modalities are fused into a single token sequence before being processed by the transformer: ECG, EDA, RESP, SpO₂, and EEG, as shown in Fig. 1(c). The 10-second windows yield $L_{\text{bio}} = 1000$ data points for peripheral signals and $L_{\text{EEG}} = 1280$ data points across $C_{\text{EEG}} = 14$ channels for EEG.

1) *Temporal alignment*: When EEG is included in the selected modality set $\mathcal{M}$, each peripheral signal $\mathbf{x}^{(m)} \in \mathbb{R}^{L_{\text{bio}}}$, $m \in \mathcal{M}_{\text{bio}}$, is resampled to L_{EEG} via linear interpolation:

$$\hat{x}_j^{(m)} = x_i^{(m)} + \left(\frac{j(L_{\text{bio}} - 1)}{L_{\text{EEG}} - 1} - i \right) (x_{i+1}^{(m)} - x_i^{(m)}), \quad (1)$$

where $i = \lfloor j(L_{\text{bio}} - 1)/(L_{\text{EEG}} - 1) \rfloor$ and $j = 0, \dots, L_{\text{EEG}} - 1$. When the EEG is absent, all signals retain their native length L_{bio} and no interpolation is applied. The interpolation step is used only to align temporal sequence lengths before channel stacking. No frequency-domain biosignal features are extracted, and the Fourier features below encode temporal position rather than signal spectra.

2) *Channel-stack fusion*: The multimodal input tensor is formed by concatenating all selected modalities along the channel dimension:

$$\mathbf{X} = [\hat{\mathbf{x}}^{(1)} \parallel \dots \parallel \hat{\mathbf{x}}^{(|\mathcal{M}|)}] \in \mathbb{R}^{C \times L}, \quad (2)$$

where $C = C_{\text{EEG}} \mathbf{1}[\text{EEG} \in \mathcal{M}] + |\mathcal{M}_{\text{bio}}|$ and $L = L_{\text{EEG}}$ if EEG is present, otherwise L_{bio} . No modality-specific branches or encoders are used; all channels are treated uniformly in a shared representation space. For the full pentamodal configuration, $C=18$ and $L=1280$, as illustrated in Fig. 1(d). The unimodal configuration corresponds to a special degenerate case of (2) with $|\mathcal{M}|=1$.

3) *Input normalization*: Prior to tokenization, each channel of $\mathbf{X}$ is normalized over the temporal dimension via a learnable Layer Normalization:

$$\text{LN}(\mathbf{x}) = \frac{\mathbf{x} - \mu_{\mathbf{x}}}{\sqrt{\sigma_{\mathbf{x}}^2 + \varepsilon}} \odot \gamma + \beta, \quad (3)$$

where $\mu_{\mathbf{x}}$ and $\sigma_{\mathbf{x}}^2$ are the mean and variance computed over the L temporal positions of that channel, and $\gamma, \beta \in \mathbb{R}^L$ are learnable affine parameters. This operation standardizes signal amplitudes across physiologically heterogeneous modalities before they enter the transformer.

4) *Tokenization*: Each temporal position $n = 1, \dots, N$ (where $N=L$) is treated as a token whose feature vector combines the raw signal values with Fourier-based positional encodings [38]. Each position is mapped to a normalized coordinate $p_n = 2(n-1)/(N-1) - 1 \in [-1, 1]$. The Fourier encoding with K frequency bands and maximum frequency $f_{\max}$ is defined as:

$$\gamma(p) = [\sin(\pi f_1 p), \cos(\pi f_1 p), \dots, \sin(\pi f_K p), \cos(\pi f_K p), p], \quad (4)$$

where the frequencies are spaced geometrically on $[1, f_{\max}/2]$:

$$f_k = 2^{(k-1) \log_2(f_{\max}/2) / (K-1)}, \quad k = 1, \dots, K. \quad (5)$$

The complete token matrix is formed by concatenating data channels and positional features at every position:

$$\mathbf{T} = [\mathbf{X}^\top \parallel \mathbf{\Gamma}] \in \mathbb{R}^{N \times C'}, \quad (6)$$

where $\mathbf{\Gamma} \in \mathbb{R}^{N \times (2K+1)}$ stacks the positional encodings row-wise and $C' = C + (2K + 1)$. With $K=6$ and $f_{\max}=10$, the positional dimension is $2K+1=13$, so $C'=C+13$. For the pentamodal case ($C=18$) this gives $\mathbf{T} \in \mathbb{R}^{1280 \times 31}$, consistent with Fig. 1(d).

C. Hierarchical Transformer Architecture

The model processes $\mathbf{T} \in \mathbb{R}^{N \times C'}$ through four sequential transformer blocks, illustrated in Fig. 1(e). Both the number of latent vectors M_ℓ and the latent dimensionality d_ℓ decrease across blocks, progressively compressing the representation towards a compact encoding for classification. The per-block hyperparameters are provided in Table VI (Appendix). The resulting model is lightweight, comprising 5.61M parameters and requiring 0.36 GFLOPs per forward pass.

1) *Latent initialization*: A global latent array $\mathbf{E} \in \mathbb{R}^{M_1 \times d_1}$ of learnable vectors is shared across the batch. At the entry to Block-1, the latent state is initialized as:

$$\mathbf{Z}^{(1)} = \mathbf{E} \in \mathbb{R}^{M_1 \times d_1}. \quad (7)$$

2) *Cross-attention*: Each block begins with a cross-attention operation in which the latent array queries the full token sequence $\mathbf{T}$, using pre-layer normalization and an additive residual connection:

$$\mathbf{Z} \leftarrow \mathbf{Z} + \text{MHA}_{\text{cr}}(\text{LN}(\mathbf{Z}), \mathbf{T}), \quad (8)$$

$$\mathbf{Z} \leftarrow \mathbf{Z} + \text{FFN}(\text{LN}(\mathbf{Z})), \quad (9)$$

where MHA_{cr} uses the latents as queries and the token matrix as keys and values, with $H_{\text{cr}}=1$ head in every block. Multi-head attention is computed as:

$$\text{MHA}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{Concat}(\text{head}_1, \dots, \text{head}_H) \mathbf{W}^O, \quad (10)$$

$$\text{head}_h = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{W}_h^Q(\mathbf{K}\mathbf{W}_h^K)^\top}{\sqrt{d_h}}\right) \mathbf{V}\mathbf{W}_h^V. \quad (11)$$

The single cross-attention head keeps the cost of reading from the token sequence proportional to $\mathcal{O}(M_\ell N)$. The feedforward network uses a GELU-activated two-layer projection:

$$\text{FFN}(\mathbf{z}) = \mathbf{W}_2 \sigma_{\text{GELU}}(\mathbf{W}_1 \mathbf{z}). \quad (12)$$

3) *Latent self-attention*: Following cross-attention, the latent array undergoes R_ℓ rounds of self-attention within the latent space at cost $\mathcal{O}(M_\ell^2)$. For each round $r = 1, \dots, R_\ell$:

$$\mathbf{Z} \leftarrow \mathbf{Z} + \text{MHA}_{\text{la}}(\text{LN}(\mathbf{Z}), \text{LN}(\mathbf{Z})), \quad (13)$$

$$\mathbf{Z} \leftarrow \mathbf{Z} + \text{FFN}(\text{LN}(\mathbf{Z})). \quad (14)$$

All attention and feedforward operations use pre-layer normalization with additive residual connections, which stabilize gradient flow across the depth of the model.

4) *Between-block transition*: After block ℓ the latent state $\mathbf{Z}_{\text{out}}^{(\ell)} \in \mathbb{R}^{M_\ell \times d_\ell}$ undergoes a two-stage transition. First, a linear projection adapts the channel dimension:

$$\tilde{\mathbf{Z}}^{(\ell)} = \mathbf{Z}_{\text{out}}^{(\ell)} \mathbf{W}_{\text{proj}}^{(\ell)} \in \mathbb{R}^{M_\ell \times d_{\ell+1}}, \quad (15)$$

where $\mathbf{W}_{\text{proj}}^{(\ell)} \in \mathbb{R}^{d_\ell \times d_{\ell+1}}$. Second, the number of latent vectors is reduced via one-dimensional adaptive average pooling along the latent axis:

$$\mathbf{Z}^{(\ell+1)} = \text{AdaptiveAvgPool}_{M_\ell \rightarrow M_{\ell+1}}(\tilde{\mathbf{Z}}^{(\ell)}) \in \mathbb{R}^{M_{\ell+1} \times d_{\ell+1}}. \quad (16)$$

This joint reduction of channel dimension and latent count enforces an information bottleneck at every block boundary, requiring each successive block to work from a more compact representation.

5) *Classification head*: After Block-4, the final latent state $\mathbf{Z}^{(4)} \in \mathbb{R}^{M_4 \times d_4}$ is averaged over latent vectors:

$$\mathbf{z}_{\text{cls}} = \frac{1}{M_4} \sum_{m=1}^{M_4} \mathbf{Z}_m^{(4)} \in \mathbb{R}^{d_4}. \quad (17)$$

A final Layer Normalization followed by a linear projection produces the class logits:

$$\hat{\mathbf{y}} = \mathbf{W}_{\text{cls}} \text{LN}(\mathbf{z}_{\text{cls}}) + \mathbf{b}_{\text{cls}} \in \mathbb{R}^{N_c}, \quad (18)$$

where $N_c = 2$ for binary workload classification.

D. Augmentation, Regularization, and Training

Three channel-synchronized augmentations are applied during training: *polarity inversion*, *additive Gaussian noise*, and *temporal masking*. Additional regularization includes label smoothing and dropout. Models are trained with *AdamW* and a cosine-annealing schedule. Full hyperparameters are provided in Table VII (Appendix).

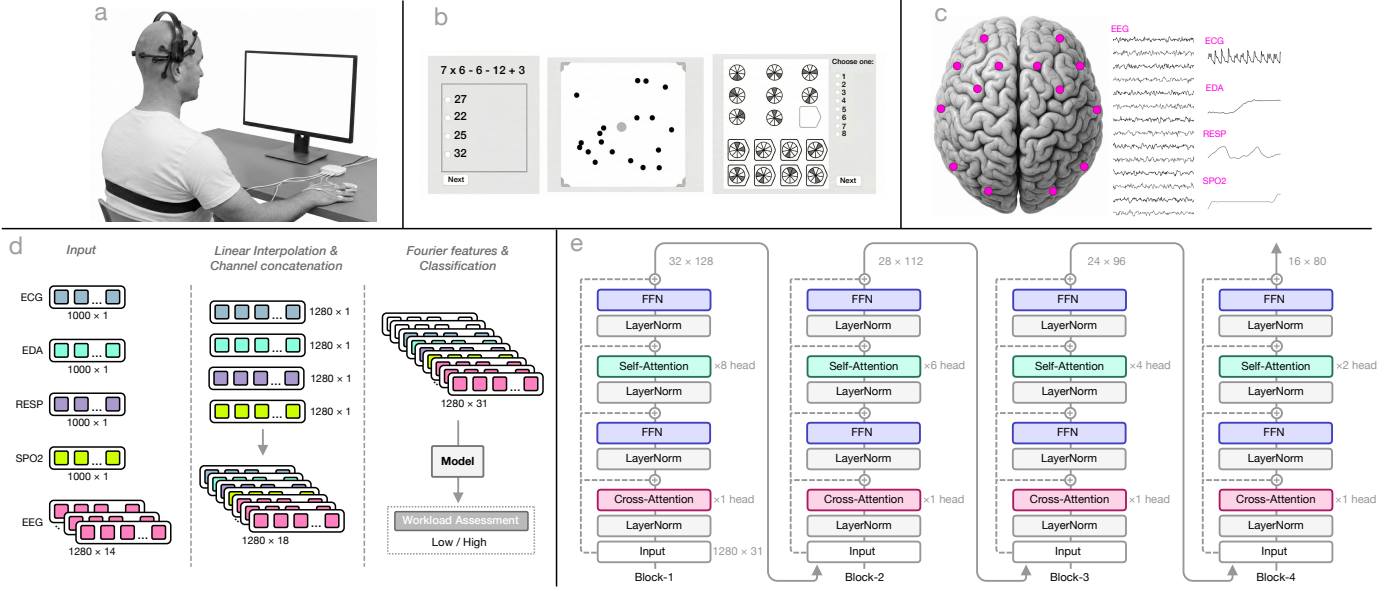


Fig. 1: Overview of the proposed multimodal cognitive workload assessment framework. **(a)** Experimental setup for physiological data acquisition. **(b)** Workload-inducing task paradigms: abstract reasoning (IQ), arithmetic problem solving (MATH), and game-based interaction (GAME), each presented under low- and high-workload conditions. **(c)** The five recorded physiological modalities (EEG: 14 channels; ECG, EDA, RESP, SpO₂: 1 channel each) with their raw signal traces. **(d)** Multimodal input construction: peripheral biosignals are linearly interpolated to the EEG temporal length (1280 samples) and concatenated along the channel axis, producing a fused tensor of 1280×18 , which expands to 1280×31 after Fourier positional features are appended. **(e)** The hierarchical four-block architecture. The fused token sequence is processed through cross-attention and repeated latent self-attention, while latent vectors and dimensionality decrease progressively across blocks: Block-1 (32×128), Block-2 (28×112), Block-3 (24×96), Block-4 (16×80), producing the final binary workload prediction (*Low/High*).

IV. EXPERIMENTAL EVALUATION & RESULTS

This section reports results for all possible modality combinations, ranging from single-modality models to the full five-modality set (ECG, EDA, RESP, SpO₂, EEG), across three tasks (IQ, MATH, GAME) and a cross-task setting (ALL), under a leave-one-subject-out (LOSO) protocol. Each task is framed as a binary classification problem using the predefined easy/hard task-difficulty conditions as controlled workload-inducing labels, rather than direct subjective NASA-TLX targets. Performance is reported using accuracy, precision, and F1, together with their arithmetic mean (*Average*), which serves as the primary criterion. *Average* provides a compact summary that favors modality combinations performing consistently across all three reported metrics.

A. Unimodal Assessment

Table I reports unimodal performance across all four task settings. EEG is the best single modality in three of four settings, achieving the highest averages of 69.22% (IQ), 72.27% (GAME), and 63.96% (ALL). For GAME, its advantage over RESP (the second-best modality) is 72.27% vs. 70.29%. In individual metrics, EEG achieves 71.97% accuracy, 73.53% precision, and 71.32% F1, while RESP reaches 69.32%, 73.99%, and 67.56%. RESP shows slightly higher precision, but EEG's higher accuracy and F1 score result in a higher

overall *Average*. This is consistent with EEG capturing task-relevant cortical dynamics of the gaming task more reliably across both workload levels.

ECG is the weakest-performing modality across all three tasks. It yields the lowest performance in the ALL setting, achieving an *Average* of 46.52%, a precision of 32.51%, failing to achieve chance-level discrimination. The low quality of this signal suggests that cardiac activity lacks the specificity needed to accurately discriminate task workloads. For MATH, the RESP yields the largest *Average* of 65.64% and slightly surpasses EEG's (65.22%). Both modalities had similar accuracy (64.77%), but RESP presented higher precision (70.35%) than EEG (67.19%), while EEG had a higher F1 (63.69%) than RESP (61.79%). These differences indicate that both modalities capture distinct characteristics of the MATH workload; RESP captures changes in overall physiological load with greater specificity, whereas EEG offers a better-balanced trade-off between precision and recall. EDA and SpO₂ represent a middle ground in terms of performance, with SpO₂ being generally more stable. All modalities showed that the worst scores were obtained in the ALL condition, since combining physiological data from three separate cognitive tasks increases inter-task variability within the same class. This pattern is consistent across all modalities.

TABLE I: Unimodal performance (*low vs high*) across all task settings. *Average*: arithmetic mean of Accuracy, Precision, and F1, used as the primary performance criterion throughout. **Bold** marks the highest *Average* within each task; underline marks the second-highest.

Task	Modality	Performance			
		Accuracy	Precision	F1	<i>Average</i>
IQ	ECG	55.68	59.53	48.12	<i>54.44</i>
IQ	EDA	63.26	68.92	60.52	<i>64.23</i>
IQ	RESP	64.39	71.75	60.77	<u>65.64</u>
IQ	SpO ₂	64.02	67.65	59.73	<i>63.80</i>
IQ	EEG	68.18	74.12	65.37	69.22
MATH	ECG	60.23	63.09	53.54	<i>58.95</i>
MATH	EDA	59.47	64.72	56.18	<i>60.12</i>
MATH	RESP	64.77	70.35	61.79	65.64
MATH	SpO ₂	62.88	67.54	60.91	<i>63.78</i>
MATH	EEG	64.77	67.19	63.69	<u>65.22</u>
GAME	ECG	60.23	67.00	54.18	<i>60.47</i>
GAME	EDA	64.02	65.29	60.32	<i>63.21</i>
GAME	RESP	69.32	73.99	67.56	<u>70.29</u>
GAME	SpO ₂	62.50	65.49	58.01	<i>61.67</i>
GAME	EEG	71.97	73.53	71.32	72.27
ALL	ECG	56.44	32.51	50.60	<i>46.52</i>
ALL	EDA	56.82	60.98	49.99	<i>55.93</i>
ALL	RESP	59.60	60.14	55.94	<u>58.56</u>
ALL	SpO ₂	55.56	57.25	52.70	<i>55.17</i>
ALL	EEG	63.76	65.52	62.61	63.96

B. Bimodal Combinations

Table II reports performance for all 10 pairwise modality combinations. EEG-containing pairs achieve the highest *Average* in most settings. For IQ, RESP+EEG obtains the best bimodal *Average* of 72.00% (71.59% accuracy, 73.49% precision, 70.93% F1), a gain of +2.78 points over the best unimodal result. The improvement is consistent across all three metrics, indicating complementarity between brain and respiratory signals for abstract reasoning. For ALL, SpO₂+EEG achieves an average of 64.00%, a marginal +0.04 improvement over unimodal EEG, with a slight F1 gain (62.97% vs. 62.61%), while accuracy and precision remain nearly flat. For MATH, ECG+EEG achieves an average of 65.85%, only +0.21 above the unimodal RESP baseline, indicating limited multimodal benefit for arithmetic workload. For GAME and ALL settings, the EEG-containing pairs perform worse than the unimodal EEG approach. The best pair, SpO₂+EEG, achieves an average of 68.59%, a decrease of −3.68 points relative to unimodal EEG alone (72.27%). SpO₂+EEG reaches 68.94% accuracy and 71.50% precision, still close to unimodal EEG, but F1 drops to 65.33% vs. 71.32%. This suggests that adding peripheral channels to the shared token sequence disrupts the EEG-based representation, rather than complementing it. The GAME task likely relies more strongly on EEG-derived information, making unimodal EEG an effective input modality. Notably, among pairs that do not include EEG, ECG+RESP achieves 65.84% for MATH, nearly matching the best EEG-containing pair, driven by strong precision (72.45%). This indicates that cardiovascular and respiratory signals in isolation

TABLE II: Bimodal performance across all task settings.

Task	Modality	Performance			
		Accuracy	Precision	F1	<i>Average</i>
IQ	ECG, EDA	63.64	64.23	60.84	<i>62.90</i>
IQ	ECG, RESP	62.88	66.84	60.60	<i>63.44</i>
IQ	ECG, SpO ₂	65.15	72.38	61.53	<i>66.35</i>
IQ	ECG, EEG	67.42	71.98	65.37	<i>68.26</i>
IQ	EDA, RESP	64.02	69.09	60.88	<i>64.66</i>
IQ	EDA, SpO ₂	65.53	69.43	63.58	<i>66.18</i>
IQ	EDA, EEG	70.45	72.10	69.79	<u>70.78</u>
IQ	RESP, SpO ₂	65.15	70.98	62.11	<i>66.08</i>
IQ	RESP, EEG	71.59	73.49	70.93	72.00
IQ	SpO ₂ , EEG	69.70	72.31	68.66	<i>70.22</i>
MATH	ECG, EDA	63.64	65.89	62.43	<i>63.99</i>
MATH	ECG, RESP	64.39	72.45	60.68	<u>65.84</u>
MATH	ECG, SpO ₂	62.12	63.81	61.11	<i>62.35</i>
MATH	ECG, EEG	64.77	70.70	62.08	65.85
MATH	EDA, RESP	64.02	70.97	61.09	<i>65.36</i>
MATH	EDA, SpO ₂	60.23	63.47	55.64	<i>59.78</i>
MATH	EDA, EEG	61.74	68.24	55.97	<i>61.98</i>
MATH	RESP, SpO ₂	65.15	67.07	64.18	<i>65.47</i>
MATH	RESP, EEG	61.36	68.00	57.42	<i>62.26</i>
MATH	SpO ₂ , EEG	60.23	69.59	54.89	<i>61.57</i>
GAME	ECG, EDA	63.26	64.97	59.80	<i>62.68</i>
GAME	ECG, RESP	65.50	71.17	62.72	<i>66.46</i>
GAME	ECG, SpO ₂	62.12	70.67	57.64	<i>63.48</i>
GAME	ECG, EEG	68.94	68.05	66.82	<i>67.94</i>
GAME	EDA, RESP	64.77	69.41	62.58	<i>65.59</i>
GAME	EDA, SpO ₂	61.74	67.16	59.03	<i>62.64</i>
GAME	EDA, EEG	66.67	65.58	62.04	<i>64.76</i>
GAME	RESP, SpO ₂	64.02	65.06	63.11	<i>64.06</i>
GAME	RESP, EEG	67.80	68.35	64.78	<u>66.98</u>
GAME	SpO ₂ , EEG	68.94	71.50	65.33	68.59
ALL	ECG, EDA	55.68	62.55	49.19	<i>55.81</i>
ALL	ECG, RESP	56.57	58.80	53.90	<i>56.42</i>
ALL	ECG, SpO ₂	55.43	54.47	49.00	<i>52.97</i>
ALL	ECG, EEG	64.02	62.73	61.95	<i>62.90</i>
ALL	EDA, RESP	56.57	61.60	52.69	<i>56.95</i>
ALL	EDA, SpO ₂	56.06	59.63	51.53	<i>55.74</i>
ALL	EDA, EEG	63.13	61.57	60.90	<i>61.87</i>
ALL	RESP, SpO ₂	56.82	62.32	52.47	<i>57.20</i>
ALL	RESP, EEG	63.01	64.89	61.98	<u>63.29</u>
ALL	SpO ₂ , EEG	63.76	65.28	62.97	64.00

carry meaningful information about arithmetic workload.

C. Trimodal Combinations

Table III presents the results for the 3-modality combinations. For IQ, the best trimodal configuration, ECG+RESP+EEG, achieves 69.10% on average, a drop of −2.90 points from the bimodal peak of 72.00%. Precision declines from 73.49% to 72.35%, and F1 drops more sharply from 70.93% to 66.77%, indicating that adding ECG to the RESP+EEG pair likely lowers recall. Regarding GAME, the best configuration is ECG+EDA+RESP, with an average of 66.46%—a combination that excludes EEG. Its metrics (65.91% accuracy, 68.16% precision, 65.32% F1) are balanced across all three dimensions, in contrast to many EEG-containing triples where precision is notably higher than F1. This suggests that the three peripheral signals produce a more consistent representation for the GAME workload. For MATH, RESP+SpO₂+EEG achieves the best result, with an average of 66.98% (66.67% accuracy, 68.89% precision, 65.38% F1), an improvement

TABLE III: Trimodal performance across all task settings.

Task	Modality	Performance			
		Accuracy	Precision	F1	Average
IQ	ECG, EDA, RESP	64.77	71.09	61.97	65.94
IQ	ECG, EDA, SpO ₂	67.80	70.29	66.80	68.30
IQ	ECG, EDA, EEG	67.05	69.03	65.76	67.28
IQ	ECG, RESP, SpO ₂	63.26	68.61	60.92	64.26
IQ	ECG, RESP, EEG	68.18	72.35	66.77	69.10
IQ	ECG, SpO ₂ , EEG	67.05	71.27	65.05	67.79
IQ	EDA, RESP, SpO ₂	66.67	70.74	64.64	67.35
IQ	EDA, RESP, EEG	67.80	69.97	64.66	67.48
IQ	EDA, SpO ₂ , EEG	67.80	72.23	65.73	68.59
IQ	RESP, SpO ₂ , EEG	68.18	69.91	67.09	68.39
MATH	ECG, EDA, RESP	64.39	70.75	61.14	65.43
MATH	ECG, EDA, SpO ₂	60.98	62.53	57.29	60.27
MATH	ECG, EDA, EEG	65.53	69.64	63.35	66.17
MATH	ECG, RESP, SpO ₂	64.02	66.52	60.07	63.54
MATH	ECG, RESP, EEG	64.02	69.36	61.31	64.90
MATH	ECG, SpO ₂ , EEG	64.02	70.60	60.71	65.11
MATH	EDA, RESP, SpO ₂	65.15	66.16	62.38	64.56
MATH	EDA, RESP, EEG	64.77	68.61	62.71	65.36
MATH	EDA, SpO ₂ , EEG	62.12	67.79	58.78	62.90
MATH	RESP, SpO ₂ , EEG	66.67	68.89	65.38	66.98
GAME	ECG, EDA, RESP	65.91	68.16	65.32	66.46
GAME	ECG, EDA, SpO ₂	62.50	69.29	58.89	63.56
GAME	ECG, EDA, EEG	65.53	68.36	64.04	65.98
GAME	ECG, RESP, SpO ₂	64.39	70.82	61.24	65.48
GAME	ECG, RESP, EEG	65.15	70.13	62.72	66.00
GAME	ECG, SpO ₂ , EEG	64.77	67.39	63.52	65.23
GAME	EDA, RESP, SpO ₂	62.88	70.39	59.44	64.24
GAME	EDA, RESP, EEG	65.91	68.84	64.27	66.34
GAME	EDA, SpO ₂ , EEG	65.91	67.89	65.22	66.34
GAME	RESP, SpO ₂ , EEG	64.39	68.09	62.30	64.93
ALL	ECG, EDA, RESP	60.10	63.04	57.76	60.30
ALL	ECG, EDA, SpO ₂	55.56	58.89	51.50	55.32
ALL	ECG, EDA, EEG	61.99	65.23	59.55	62.26
ALL	ECG, RESP, SpO ₂	55.68	59.52	51.69	55.63
ALL	ECG, RESP, EEG	62.63	64.52	61.29	62.81
ALL	ECG, SpO ₂ , EEG	63.64	64.83	62.77	63.75
ALL	EDA, RESP, SpO ₂	58.96	57.72	56.50	57.73
ALL	EDA, RESP, EEG	64.02	64.98	63.52	64.17
ALL	EDA, SpO ₂ , EEG	62.88	67.93	60.01	63.61
ALL	RESP, SpO ₂ , EEG	61.99	63.01	61.38	62.13

of +1.13 over bimodal, with gains across all three metrics. For ALL, EDA+RESP+EEG reaches 64.17% (+0.17 over bimodal), with accuracy, precision, and F1 closely aligned at 64.02%, 64.98%, and 63.52% respectively.

D. Quadmodal Combinations

Table IV reports performance for the 4-modality combinations. For MATH, the best result is obtained by ECG+EDA+RESP+SpO₂, with an average of 64.89%, the only combination without EEG to rank first at any level across the entire evaluation. Its metrics (64.02% accuracy, 69.54% precision, 61.10% F1) follow the characteristic MATH profile, with relatively high precision and lower F1. All four EEG-containing quadmodal configurations score below this, ranging from 60.60% to 63.24%. At the unimodal, trimodal, and quadmodal levels, EEG does not improve upon physiological-only combinations for MATH, and its inclusion appears to degrade performance for this task. For IQ, the best quadmodal result is ECG+EDA+RESP+EEG at an average of 68.03%,

TABLE IV: Quadmodal performance across all task settings.

Task	Modality	Performance			
		Accuracy	Precision	F1	Average
IQ	ECG, EDA, RESP, SpO ₂	66.29	72.33	64.07	67.56
IQ	ECG, EDA, RESP, EEG	67.42	70.75	65.92	68.03
IQ	ECG, EDA, SpO ₂ , EEG	66.67	68.57	65.69	66.98
IQ	ECG, RESP, SpO ₂ , EEG	66.29	70.58	64.30	67.06
IQ	EDA, RESP, SpO ₂ , EEG	65.53	69.16	63.30	66.00
MATH	ECG, EDA, RESP, SpO ₂	64.02	69.54	61.10	64.89
MATH	ECG, EDA, RESP, EEG	60.23	67.67	55.91	61.27
MATH	ECG, EDA, SpO ₂ , EEG	59.09	69.05	53.65	60.60
MATH	ECG, RESP, SpO ₂ , EEG	61.74	70.10	57.87	63.24
MATH	EDA, RESP, SpO ₂ , EEG	59.85	69.23	54.74	61.27
GAME	ECG, EDA, RESP, SpO ₂	62.88	62.08	58.29	61.08
GAME	ECG, EDA, RESP, EEG	66.29	65.32	64.14	65.25
GAME	ECG, EDA, SpO ₂ , EEG	67.42	71.45	65.91	68.26
GAME	ECG, RESP, SpO ₂ , EEG	68.18	71.83	66.41	68.81
GAME	EDA, RESP, SpO ₂ , EEG	66.29	68.83	64.57	66.56
ALL	ECG, EDA, RESP, SpO ₂	57.83	60.19	55.30	57.77
ALL	ECG, EDA, RESP, EEG	57.32	61.09	53.73	57.38
ALL	ECG, EDA, SpO ₂ , EEG	60.35	59.61	58.01	59.32
ALL	ECG, RESP, SpO ₂ , EEG	61.24	61.95	60.70	61.30
ALL	EDA, RESP, SpO ₂ , EEG	57.83	60.99	54.91	57.91

which is 3.97 points below the bimodal peak of 72.00%. Precision falls from 73.49% to 70.75% and F1 from 70.93% to 65.92%, indicating that adding ECG and EDA to the RESP+EEG pair disrupts the precision-recall balance for abstract reasoning.

E. Pentamodal Combination

Table V reports results for all 5-modality combinations. The full pentamodal configuration achieves the highest Average across the entire evaluation: 73.02% for IQ, with all three individual metrics elevated (72.35% accuracy, 76.27% precision, 70.43% F1). This is the only configuration across all 31 combinations in which precision exceeds 76%, suggesting that the full signal set enables more balanced classification across both workload levels. The gain over the best bimodal result (72.00%) is +1.02 points, driven primarily by a jump in precision from 73.49% to 76.27%. For GAME, the pentamodal result is 71.15% (70.83% accuracy, 72.62% precision, 69.99% F1), below the unimodal EEG result of 72.27%, confirming that no combination surpasses EEG alone for this task. For MATH (65.15%) and ALL (63.01%), differences from the quadmodal bests are small (+0.26 and +1.71 points), consistent with the limited sensitivity of these tasks to the number of modalities included. The pentamodal configuration that yields the best cross-task results, despite intermediate combinations often performing worse than unimodal EEG, suggests that the benefit of the full signal set is a property of the complete combination rather than a cumulative effect of adding modalities one by one. Partial combinations may introduce less effective representations in the shared latent space, where dominant channels are diluted without sufficient complementary information from the others.

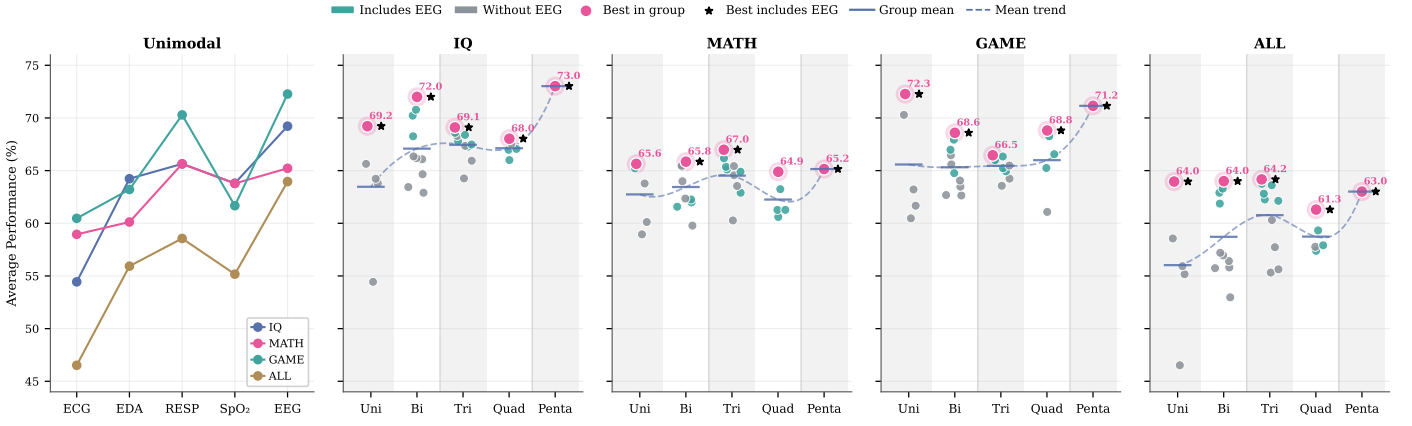


Fig. 2: Per-task *Average* performance (arithmetic mean of Accuracy, Precision, and F1) across all 31 modality combinations. **Left panel:** unimodal *Average* per modality for each task setting (IQ, MATH, GAME, ALL). **Panels 2–5:** strip plots of all combinations from unimodal through pentamodal, one panel per task. Each dot represents one combination: teal dots include EEG, grey dots do not. Pink dots mark the best-performing combination per level; ★ indicates that the best combination includes EEG. Horizontal bars denote the group means; the dashed line shows the mean trend across levels.

TABLE V: Pentamodal performance across all task settings.

Task	Modality	Performance			
		Accuracy	Precision	F1	<i>Average</i>
IQ	ECG, EDA, RESP, SpO ₂ , EEG	72.35	76.27	70.43	73.02
MATH	ECG, EDA, RESP, SpO ₂ , EEG	64.39	69.00	62.06	<i>65.15</i>
GAME	ECG, EDA, RESP, SpO ₂ , EEG	70.83	72.62	69.99	<i>71.15</i>
ALL	ECG, EDA, RESP, SpO ₂ , EEG	62.88	64.18	61.98	<i>63.01</i>

cohort size. Overall, the results indicate that EEG is the most reliable single modality, while the proposed unified framework can exploit all modalities when complementary information is available. The channel-stack fusion design also reduces model size and inference time compared with late-fusion alternatives, supporting its suitability for resource-constrained workload assessment.

V. CONCLUSION

F. Overall Analysis & Discussion

Figure 2 summarizes the main trends across all 31 modality combinations. EEG is the strongest single modality, achieving the highest *Average* in IQ, GAME, and ALL, while RESP performs slightly better for MATH. The ALL setting consistently produces lower scores, indicating that pooling different task types increases within-class variability and makes cross-task workload recognition more difficult. The effect of multimodal fusion is task-dependent. For IQ, the full pentamodal configuration achieves the highest score across the entire evaluation (73.02%). For GAME, however, unimodal EEG remains the best configuration, suggesting that additional peripheral signals may dilute rather than improve the task-relevant EEG representation. For MATH, EEG and non-EEG combinations perform similarly, indicating limited added value from brain activity for this task. These findings show that adding modalities does not guarantee improved performance and that the optimal sensor set depends on the cognitive task. When averaged across all four settings, the full pentamodal configuration yields the highest cross-task *Average* (68.08%), closely followed by unimodal EEG (67.67%). Thus, the pentamodal model provides the best overall descriptive performance, but the gain over EEG alone is small. Wilcoxon tests further suggest that the advantage of the full set of modalities should be interpreted with caution, given the small

This pilot study presented a unified, modality-agnostic hierarchical transformer framework for cognitive workload assessment and evaluated it across all possible combinations of five physiological modalities: ECG, EDA, RESP, SpO₂, and EEG, using a leave-one-subject-out protocol. Overall, the results show that EEG is the strongest single modality in most settings. Adding more modalities does not consistently improve performance, as intermediate combinations often introduce interference. The full five-modality configuration achieves the highest cross-task average (68.08%) and the highest single-task average on the abstract reasoning task (73.02%), demonstrating that the proposed unified framework can leverage complementary physiological signals without task-specific design. Beyond performance, the channel stack fusion approach requires approximately half as many parameters as embedding-level late-fusion alternatives and consistently achieves lower inference latency across all modalities, making it a practical choice for resource-constrained deployments. The small cohort ($n=11$) and single dataset scope reflect the pilot nature of the study; broader validation across larger and more diverse populations, targeted ablations of the latent bottleneck, Fourier encoding, and cross-attention design, and subjective workload modeling using NASA-TLX remain open directions for future work.

ETHICAL IMPACT STATEMENT

The experimental procedures were approved by the Institutional Review Board. Written informed consent was obtained from all participants prior to the experiment, and no personal information was recorded. The study involved 11 participants, including 6 males and 5 females, aged 20 to 40 years, with a mean age of 25 ± 5.5 years. Participants did not report neurological conditions and were not under the influence of substances known to impair nervous system function, such as alcohol or nicotine, as disclosed on the day of the experiment. This study presents a framework for cognitive workload assessment from physiological signals, with potential applications in adaptive human-machine systems and safety-critical environments. The cohort size, controlled laboratory setting, consumer-grade EEG device, and single-dataset protocol limit generalizability to broader populations and real-world settings. Binary easy/hard labels reflect controlled task-difficulty conditions rather than direct subjective workload labels. NASA-TLX scores were collected as subjective workload verification, but their distributions and correlations with model outputs were not analyzed in this study. The reported results should therefore be interpreted as preliminary. Individual physiological variability, sensor placement, recording quality, interpolation-based temporal alignment, and environmental factors may affect performance in practice. Deployment in applied settings requires validation in larger, more diverse populations, across varied experimental protocols, and through prospective studies assessing reliability across demographic groups. Automated workload assessment also carries risks in high-stakes decision-making without human oversight, since misclassification could lead to inappropriate interventions or failure to respond to elevated cognitive demand. Any future deployment should involve domain experts, ethicists, and end users in the design process.

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APPENDIX

A. Details & Complementary Experiments

Figure 3 reports the cross-task *Average* performance for all 31 modality combinations, with each entry averaged over IQ, MATH, GAME, and ALL. Tables VI and VII report the per-block architecture hyperparameters and the full training configuration used across all experiments. Table VIII provides the corresponding numerical results for Figure 3, extending the main results by reporting the cross-task *Average* for every modality combination. Figure 4 and Table IX compare the

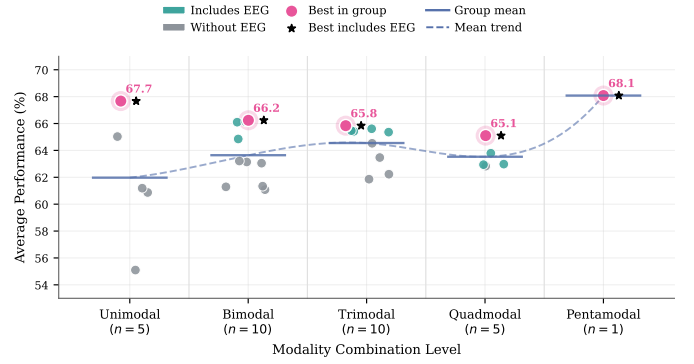


Fig. 3: Cross-task *Average* performance for all 31 modality combinations, with each entry averaged over IQ, MATH, GAME, and ALL. Full numerical results are in Table VIII (Appendix).

proposed channel-stack fusion against two standard late-fusion strategies, namely element-wise addition and concatenation of modality embeddings, across bimodal through pentamodal configurations. Results in Table IX are reported for the best-performing modality combination at each level. A key structural difference is that late-fusion methods require two separate model instances, one for the multichannel EEG and one for the single-channel peripheral signals, whereas the proposed method processes all modalities through a single shared architecture by stacking them along the channel dimension. This design difference directly affects computational and inference costs: the proposed method requires approximately half as many parameters (5.6M vs. 11.2M) and consistently achieves lower latency across all modality levels, as visualized in Figure 4. Table X reports the results of the Wilcoxon signed-rank test comparing EEG and RESP across the four task settings. Table XI provides an additional comparison with previous EEG-only methods evaluated on the same pilot dataset and LOSO protocol, including *EEGNet* and *1BT*.

TABLE VI: Per-block architecture hyperparameters.

Parameter	Value
Block 1: latent vectors M_1 , dim d_1	32, 128
Block 2: latent vectors M_2 , dim d_2	28, 112
Block 3: latent vectors M_3 , dim d_3	24, 96
Block 4: latent vectors M_4 , dim d_4	16, 80
Cross-attention heads H_{cr}	1
Cross-attention head dim d_{cr}	64, 48, 32, 16
Self-attention heads H_{la}	8, 6, 4, 2
Self-attention head dim d_{la}	64, 48, 32, 16
Self-attention rounds per block R	8, 6, 4, 2
Fourier frequency bands K	6
Maximum frequency f_{max}	10

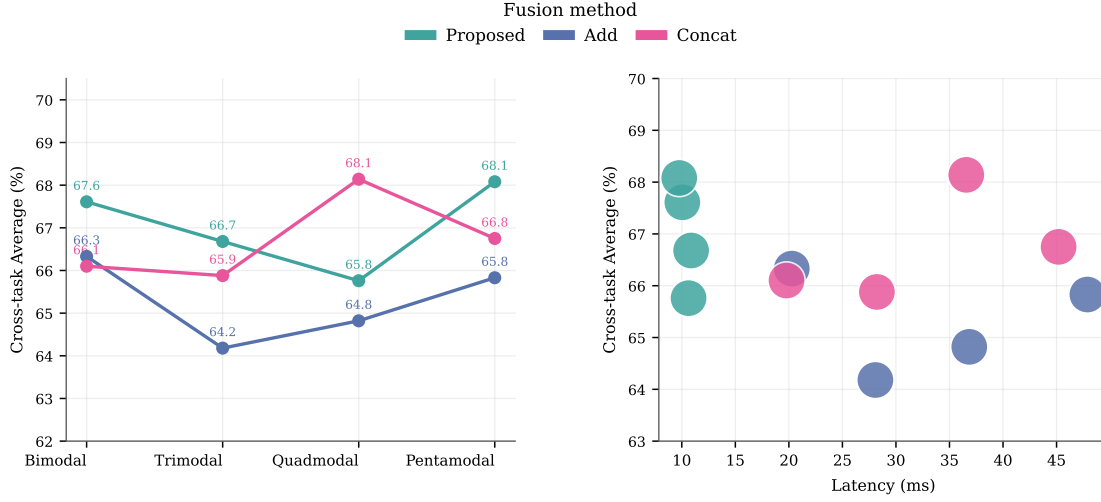


Fig. 4: Comparison of the proposed channel-stack fusion against late-fusion alternatives (add, concat). **Left:** cross-task Average per modality level. **Right:** cross-task Average versus inference latency. Values correspond to Table IX.

TABLE VII: Augmentation, regularization, and training configuration.

Parameter	Value
<i>Polarity inversion</i>	$p \sim \mathcal{U}(0, 0.05)$
<i>Additive noise</i>	$p \sim \mathcal{U}(0, 0.05), \alpha \sim \mathcal{U}(1, 1000)$
<i>Temporal masking</i>	$p \sim \mathcal{U}(0, 0.05), f \sim \mathcal{U}(0.15, 0.30)$
<i>Label smoothing ϵ_{ls}</i>	0.10
<i>Attention dropout</i>	0.10
<i>FF dropout</i>	0.10
Optimizer	AdamW
Learning rate η	1×10^{-4}
LR schedule	Cosine annealing
Weight decay λ	0.05
Warm-up epochs	20
Main epochs	170
Cooldown epochs	10
Total epochs	200
Batch size	32

TABLE VIII: Average Performance per modality or combination of modalities across the 4 tasks.

Modality	Performance			
	Accuracy	Precision	F1	Average
ECG	58.15 $\pm$ 5.20	55.53 $\pm$ 4.20	51.61 $\pm$ 10.16	55.10
EDA	60.89 $\pm$ 5.60	64.98 $\pm$ 10.41	56.75 $\pm$ 8.82	60.87
RESP	64.52 $\pm$ 6.43	69.06 $\pm$ 8.04	61.52 $\pm$ 9.51	65.03
SpO ₂	61.24 $\pm$ 6.42	64.48 $\pm$ 10.56	57.84 $\pm$ 9.03	61.19
EEG	67.17 $\pm$ 6.66	70.09 $\pm$ 6.42	65.75 $\pm$ 8.02	67.67
ECG, EDA	61.56 $\pm$ 6.16	64.41 $\pm$ 10.80	58.06 $\pm$ 9.39	61.34
ECG, RESP	62.34 $\pm$ 6.08	67.32 $\pm$ 6.41	59.48 $\pm$ 8.45	63.05
ECG, SpO ₂	61.21 $\pm$ 5.44	65.33 $\pm$ 8.92	57.32 $\pm$ 8.68	61.29
ECG, EEG	66.29 $\pm$ 8.54	68.37 $\pm$ 11.63	64.05 $\pm$ 11.14	66.24
EDA, RESP	62.35 $\pm$ 5.81	67.77 $\pm$ 7.29	59.31 $\pm$ 8.31	63.14
EDA, SpO ₂	60.89 $\pm$ 4.36	64.92 $\pm$ 7.74	57.44 $\pm$ 7.86	61.08
EDA, EEG	65.50 $\pm$ 8.79	66.87 $\pm$ 14.23	62.18 $\pm$ 12.28	64.85
RESP, SpO ₂	62.79 $\pm$ 6.28	66.36 $\pm$ 7.00	60.47 $\pm$ 8.33	63.21
RESP, EEG	65.94 $\pm$ 7.77	68.68 $\pm$ 9.68	63.78 $\pm$ 9.44	66.13
SpO ₂ , EEG	65.66 $\pm$ 8.15	69.67 $\pm$ 9.83	62.96 $\pm$ 10.39	66.10
ECG, EDA, RESP	63.79 $\pm$ 5.42	68.26 $\pm$ 6.71	61.55 $\pm$ 7.11	64.53
ECG, EDA, SpO ₂	61.71 $\pm$ 5.31	65.25 $\pm$ 7.99	58.62 $\pm$ 7.84	61.86
ECG, EDA, EEG	65.02 $\pm$ 5.93	68.07 $\pm$ 5.61	63.18 $\pm$ 8.58	65.42
ECG, RESP, SpO ₂	61.84 $\pm$ 6.91	66.37 $\pm$ 9.78	58.48 $\pm$ 10.10	62.23
ECG, RESP, EEG	65.00 $\pm$ 5.88	69.09 $\pm$ 6.98	63.02 $\pm$ 7.29	65.70
ECG, SpO ₂ , EEG	64.87 $\pm$ 6.47	68.52 $\pm$ 7.18	63.01 $\pm$ 7.82	65.47
EDA, RESP, SpO ₂	63.42 $\pm$ 6.19	66.25 $\pm$ 10.00	60.74 $\pm$ 9.12	63.47
EDA, RESP, EEG	65.62 $\pm$ 6.18	68.10 $\pm$ 8.07	63.79 $\pm$ 8.52	65.84
EDA, SpO ₂ , EEG	64.68 $\pm$ 5.87	68.96 $\pm$ 6.74	62.44 $\pm$ 7.69	65.36
RESP, SpO ₂ , EEG	65.31 $\pm$ 6.41	67.48 $\pm$ 6.53	64.04 $\pm$ 7.41	65.61
ECG, EDA, RESP, SpO ₂	62.76 $\pm$ 5.85	66.04 $\pm$ 9.25	59.69 $\pm$ 8.56	62.83
ECG, EDA, RESP, EEG	62.82 $\pm$ 5.31	66.21 $\pm$ 8.71	59.93 $\pm$ 8.18	62.98
ECG, EDA, SpO ₂ , EEG	63.36 $\pm$ 4.30	67.17 $\pm$ 7.42	60.82 $\pm$ 6.73	63.79
ECG, RESP, SpO ₂ , EEG	64.36 $\pm$ 6.16	68.62 $\pm$ 6.98	62.32 $\pm$ 7.78	65.10
EDA, RESP, SpO ₂ , EEG	62.38 $\pm$ 6.62	67.05 $\pm$ 7.69	59.38 $\pm$ 9.21	62.94
ECG, EDA, RESP, SpO ₂ , EEG	67.61 $\pm$ 7.52	70.52 $\pm$ 7.00	66.12 $\pm$ 9.26	68.08

TABLE IX: Cross-task *Average* performance, computational cost, and inference cost of late-fusion methods (add, concat) versus the proposed channel-stack fusion, across modality levels. For each level and task, the best-performing modality combination is selected, and its *Average* (arithmetic mean of Accuracy, Precision, and F1) is reported. Cross-task *Average* is the arithmetic mean across the four task settings (IQ, MATH, GAME, ALL). Computational and inference costs are measured on an NVIDIA RTX 4090 GPU with batch size 1.

Modality Level	Fusion	Task <i>Average</i>				Cross-task <i>Average</i>	Computational Cost		Inference Cost		
		IQ	MATH	GAME	ALL		Params (M)	GFLOPs	Latency (ms)	GPU↓	Samples/s GPU↑
Bimodal	proposed	72.00	65.85	68.59	64.00	67.61	5.60	0.35	10.05		99.52
	add	69.60	66.54	65.27	63.91	66.33	11.20	0.69	20.27		49.33
	concat	68.24	64.25	69.15	62.76	66.10	11.20	0.69	19.78		50.55
Trimodal	proposed	69.10	66.98	66.46	64.17	66.68	5.60	0.35	10.87		92.00
	add	68.10	62.52	61.61	64.47	64.18	11.20	1.03	28.08		35.61
	concat	65.47	69.37	62.84	65.82	65.88	11.20	1.03	28.22		35.43
Quadmodal	proposed	68.03	64.89	68.81	61.30	65.76	5.60	0.35	10.63		64.08
	add	66.97	61.89	66.38	64.02	64.82	11.20	1.37	36.85		27.14
	concat	71.66	68.54	65.42	66.92	68.14	11.20	1.37	36.57		27.35
Pentamodal	proposed	73.02	65.15	71.15	63.01	68.08	5.61	0.36	9.76		102.48
	add	68.15	62.76	65.84	66.57	65.83	11.20	1.71	47.88		20.89
	concat	66.86	66.10	67.77	66.26	66.75	11.20	1.71	45.19		22.13

Per-task values are the *Average* (arithmetic mean of Accuracy, Precision, F1) for the best-performing modality combination at each level, taken from the main results tables (proposed) and the fusion table (add/concat). For the late-fusion methods, Params and GFLOPs reflect the combined cost of both model instances. Latency is the mean per-batch inference time; Samples/s is the throughput. ↓ lower is better; ↑ higher is better.

TABLE X: Wilcoxon signed-rank test: EEG vs. RESP.

Task	EEG	RESP	W	p	r	Significance
IQ	68.18	64.39	9.0	0.109	0.80	—
MATH	64.77	64.77	21.5	0.905	0.52	—
GAME	71.97	69.32	25.5	0.838	0.54	—
ALL	63.76	59.60	6.0	0.014	0.91	✓

Accuracy (%) values are taken from Table I. r : rank-biserial correlation (effect size).

TABLE XI: Comparison with previous EEG-only methods evaluated on the same pilot dataset using the LOSO protocol.

Method	Modality	Params	GFLOPs	Mean Avg.
EEGNet [39]	EEG	0.01M	0.02	64.47
IBT [19]	EEG	0.45M	0.02	69.37
Proposed	ECG, EDA, RESP, SpO ₂ , EEG	5.61M	0.36	69.77

Mean Avg. is computed only over IQ, MATH, and GAME, since previous EEG-only methods do not report the pooled ALL setting.